Concepção e Autoria Multimédia

Carlos Vaz de Carvalho

Project Development Phases

- Specification and design
  - Specification
  - Design
  - Proof of Concept



Project Development Phases

- Specification and Design
  - Specifying a product involves **listing everything you require from the product**. During specification you do not need to specify how the individual features will work – that will happen during the design phase – but you need to have a framework of **features and requirements** on which to base your design.
  - The design phase takes the specifications and expands them to produce a design document detailed enough for programmers and media developers to work from. The design document often includes a **'storyboard'** or 'paper prototype' that can be used to evaluate the success of the specification and design phase.
  - This phase often ends with the development of a **proof of concept** that is used to validate and evaluate the design.
  - In other models of multimedia project management, specification and design are viewed as two distinct phases. There are often considerable **iterations between these stages**.



Project Development Phases

- Specification and Design - Specification
  - A full **list of functions intended for the package** is compiled and documented using the information gathered in the needs analysis and as agreed with the client. You can list all the functional requirements in a numbered list, with clear identification of those **functions that are mandatory requirements and those that are desirable** but not mandatory.
  - The next step is to list the ways in which different components and functions in the package will **interact**.
  - An overview of the **structure and navigation** is then derived from an analysis of the content requirements, the functionality and interaction requirements.

Project Development Phases

- Specification and Design - Specification
  - The **graphical treatment** is developed by the graphic designer with consideration for requirement issues (e.g. target user demographic, content, purpose of product) and technical constraints (e.g. file-size efficiency issues, target environment, costing issues, etc.).
  - The **packaging and distribution** requirements are specified and documented. The provider/developer of the packaging is also determined.

Project Development Phases

- Specification and Design - Specification
  - Your specification statement should include:
    - a **list of all the functions** that you expect the package to be able to handle
    - **interaction requirements** between all the components
    - a definition of a **navigation system** that best accommodates the content
    - a statement of the **desired graphical style** you would like the package to have
    - a description of the kind of **Help to be provided** (on-line versus printed documentation, task oriented versus interface oriented, technical support desk or email addresses to which users can address problems)
    - a list of necessary **support materials**
    - a statement of replication and **distribution requirements**.

Project Development Phases

- Specification and Design - Specification - Functionality Requirements
  - This list will be derived from your needs analysis and content scope. For instance, you may need to specify whether users will be able to cut and paste text into a notebook, whether there should be a map of the content, on-line Help, etc.
    - security requirements (if you need log-ins, passwords, etc.)
    - print options (Can the users print individual screens? The whole package? Some items only?)
    - subject specific requirements (like display of mathematical symbols, tables, voice recognition, etc.)
    - search options
    - help and technical support documentation (Will it be provided as part of the package or as a print document? What form will the help take? Will there be an on-line tutorial for first-time users?)
  - Every additional feature affects package performance, simplicity of navigation, package maintenance and support costs as well as the development timeline and cost. Start the list with the most obvious and necessary functions for the package. You can then add to the list, if necessary.
  - Any modifications to these specifications, and their reasons, should be documented and distributed to the whole team.

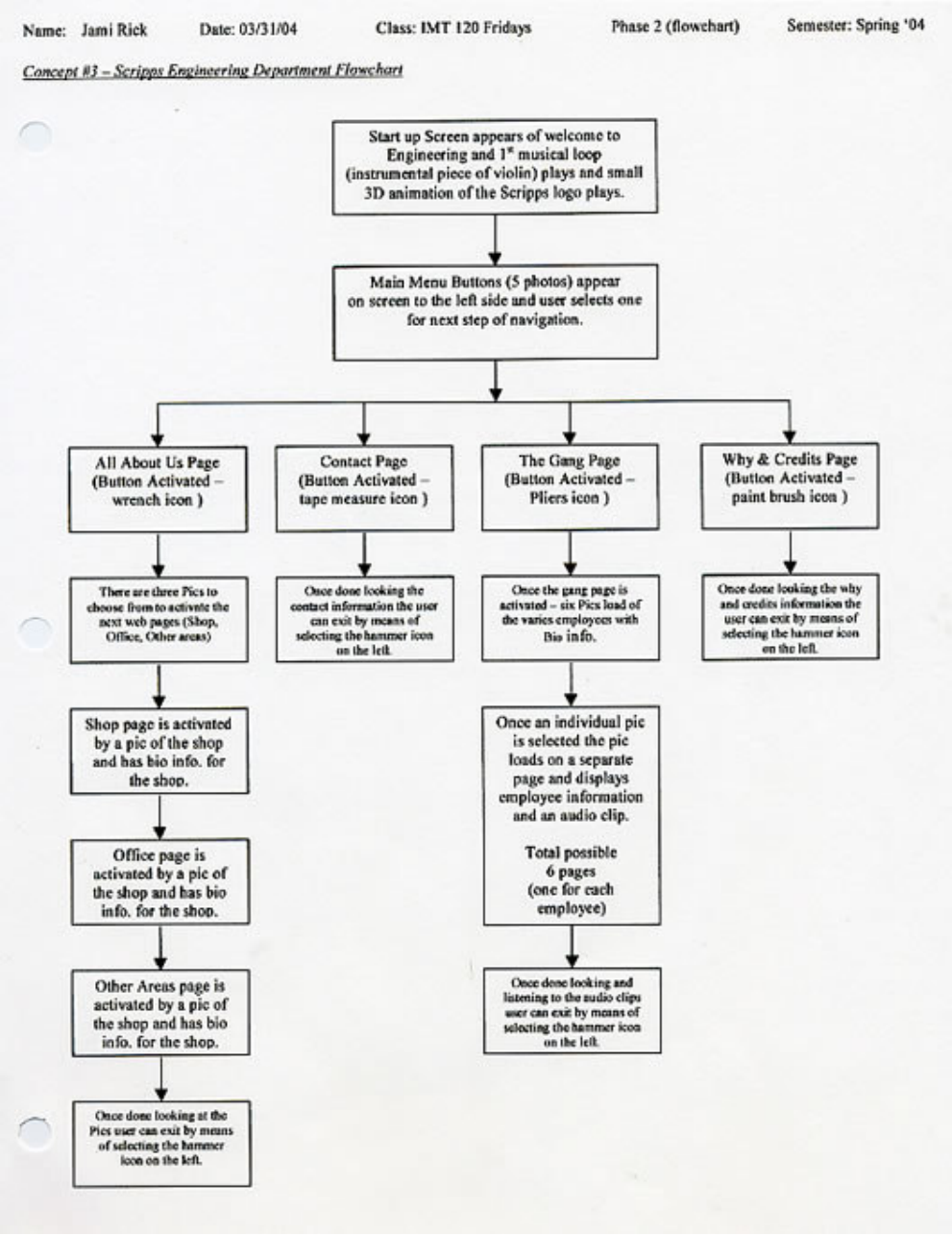
Project Development Phases

- Specification and Design - Specification - Interaction Requirements
  - The **interaction requirements** describe the way in which the different components and functions in the package will interact. They will further clarify how the package is going to work and how users will find their way through the package. The requirements will be iteratively defined along with the navigation overview.
  - As an example, the following sorts of things would be listed separately:
    - users need to be able to access the clipboard from every screen in the package
    - if users need to be able to cut and paste terms into a notebook, or type their own entries.
  - You will probably find that as the design takes place you may need to modify these specifications. Any modifications, and their reasons, should be documented and distributed to the whole team.

Project Development Phases

- Specification and Design - Specification - Navigation Overview
  - An overview of the **structure and navigation** of your package can be developed once you have determined the functionality and interaction requirements discussed above. You will need to decide on a **navigation system** that best accommodates the content.
  - You may wish to **base the navigation on a metaphor** of some kind (for example an office, a work of art, a briefcase, a board game). However, careful thought must be put into the logic and consistency of any metaphor, and it should not adversely affect the clarity of the navigation system.
  - You also need to decide how to incorporate all the media you anticipate will be included in the package.
  - Any changes that occur to this navigation overview in the design phase should be documented, with reasons, and distributed to the whole team.

Project D


Name: Jami Rick
Date: 03/31/04
Class: IMT 120 Fridays
Phase 2 (flowchart)
Semester: Spring '04
Concept #3 – Scripps Engineering Department Flowchart
Start up Screen appears of welcome to Engineering and 1st musical loop (instrumental piece of violin) plays and small 3D animation of the Scripps logo plays.
Main Menu Buttons (5 photos) appear on screen to the left side and user selects one for next step of navigation.
All About Us Page (Button Activated – wrench icon)
Contact Page (Button Activated – tape measure icon)
The Gang Page (Button Activated – Pliers icon)
Why & Credits Page (Button Activated – paint brush icon)
There are three Pics to choose from to activate the next web pages (Shop, Office, Other areas)
Once done looking the contact information the user can exit by means of selecting the hammer icon on the left.
Once the gang page is activated – six Pics load of the varies employees with Bio info.
Once done looking the why and credits information the user can exit by means of selecting the hammer icon on the left.
Shop page is activated by a pic of the shop and has bio info. for the shop.
Office page is activated by a pic of the shop and has bio info. for the shop.
Other Areas page is activated by a pic of the shop and has bio info. for the shop.
Once done looking at the Pics user can exit by means of selecting the hammer icon on the left.
Once an individual pic is selected the pic loads on a separate page and displays employee information and an audio clip.
Total possible 6 pages (one for each employee)
Once done looking and listening to the audio clips user can exit by means of selecting the hammer icon on the left.

Project Development Phases

- Specification and Design - Specification - Support Materials
  - Content and design requirements of all support materials. Support materials could include:
    - videotapes / audiotapes
    - web sites/CD-ROMs
    - documentation and print-based help
    - materials for practical manipulation.
  - Workshops, seminars or briefings you may need to support the package.
  - During the development of the scope document you will have determined the kinds of support materials that you will need. You will probably also establish who will be responsible for the production of those materials.
  - The project manager will be responsible for coordinating the development of the support materials in parallel to the development of the package.

Project Development Phases

- Specification and Design - Specification - Packaging and Distribution
  - It will be necessary at this stage to specify the replication and distribution requirements of the package, e.g. the numbers of CDs to be replicated and the type of replication required. Specifying these details now will allow the project manager to establish timelines for replication and distribution of the package.
  - You need to ensure that the design and packaging of the product complies with the relevant standards such as Corporate Guidelines and any style guides set by your department or faculty.
  - If the package is a CD-ROM it may require no other packaging than the standard CD case or slip cover. Software licensing agreements (i.e. the use of QuickTime or Macromedia logos, etc) may play a part in determining the final packaging for the CD-ROM.
  - If you are supplying supporting materials with the CD, you will need to consider how these support materials will be provided and whether there needs to be some kind of unifying design and packaging. You will also need to consider how the packing will affect the method and cost of distribution.

Project Development Phases

- Specification and Design - Design
  - Designing the product requires the project team to take the product specifications and expand them into a **design document** detailed enough for media producers and programmers to use in developing and integrating the different components into a completed product. This involves designing the detailed navigation and interaction, the product interface and the media elements. The programming tools and methodology should also be determined and documented in the design document.
  - The design document often includes a ‘storyboard’ or ‘paper prototype'. These may include the mock sample screens as well as a flowchart to map the structure and navigation. The storyboard and flowchart can be used to evaluate the design by tracing each specification and confirming that it has been effectively transferred into the design.

Objective #_____ Storyboard # 1 of 43

Understanding Your Automobile



Card Info

Card Name: Title
Card Number:
No. of Buttons: 1
No. of Fields:

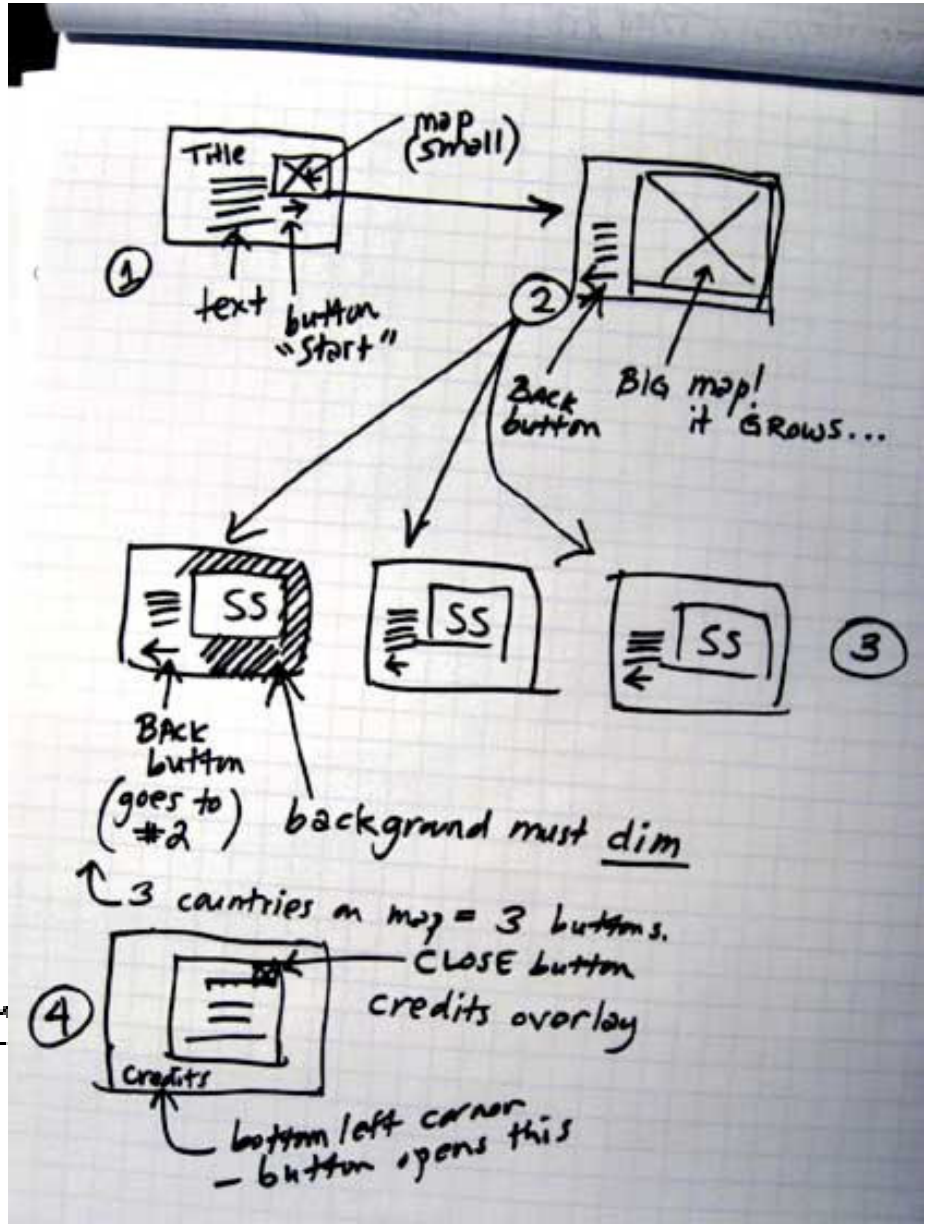
Background Name:

Button Info

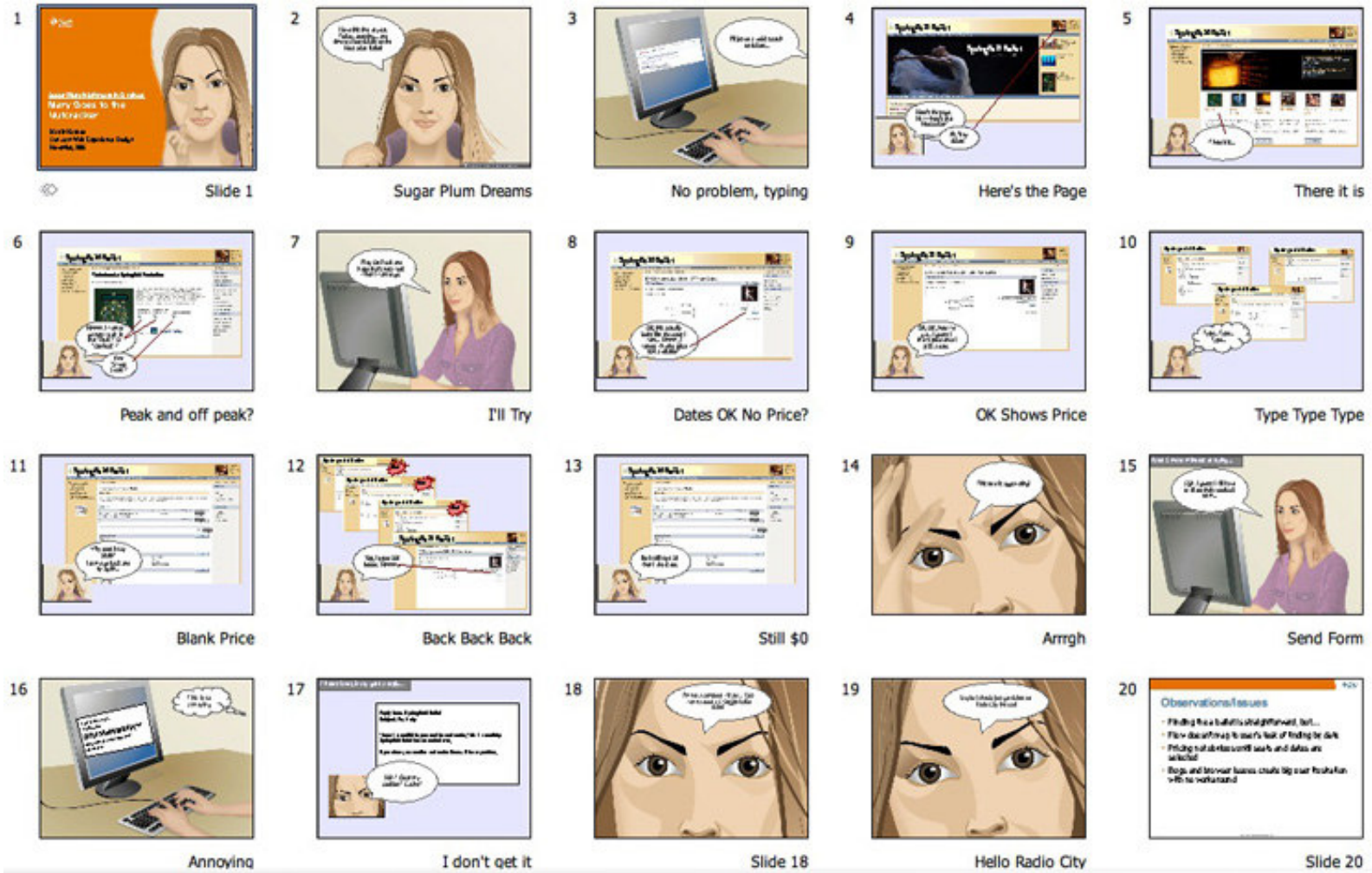
If	then
Right arrow	go next

Comments: Background color: Sky Blue
Text: Heart Yearns up: bold Bright Yellow
Graphic: Red sports-touring car
Programming: The car will drive across the screen left to right as the screen when it leaves the right side the screen will automatically go to the next screen.

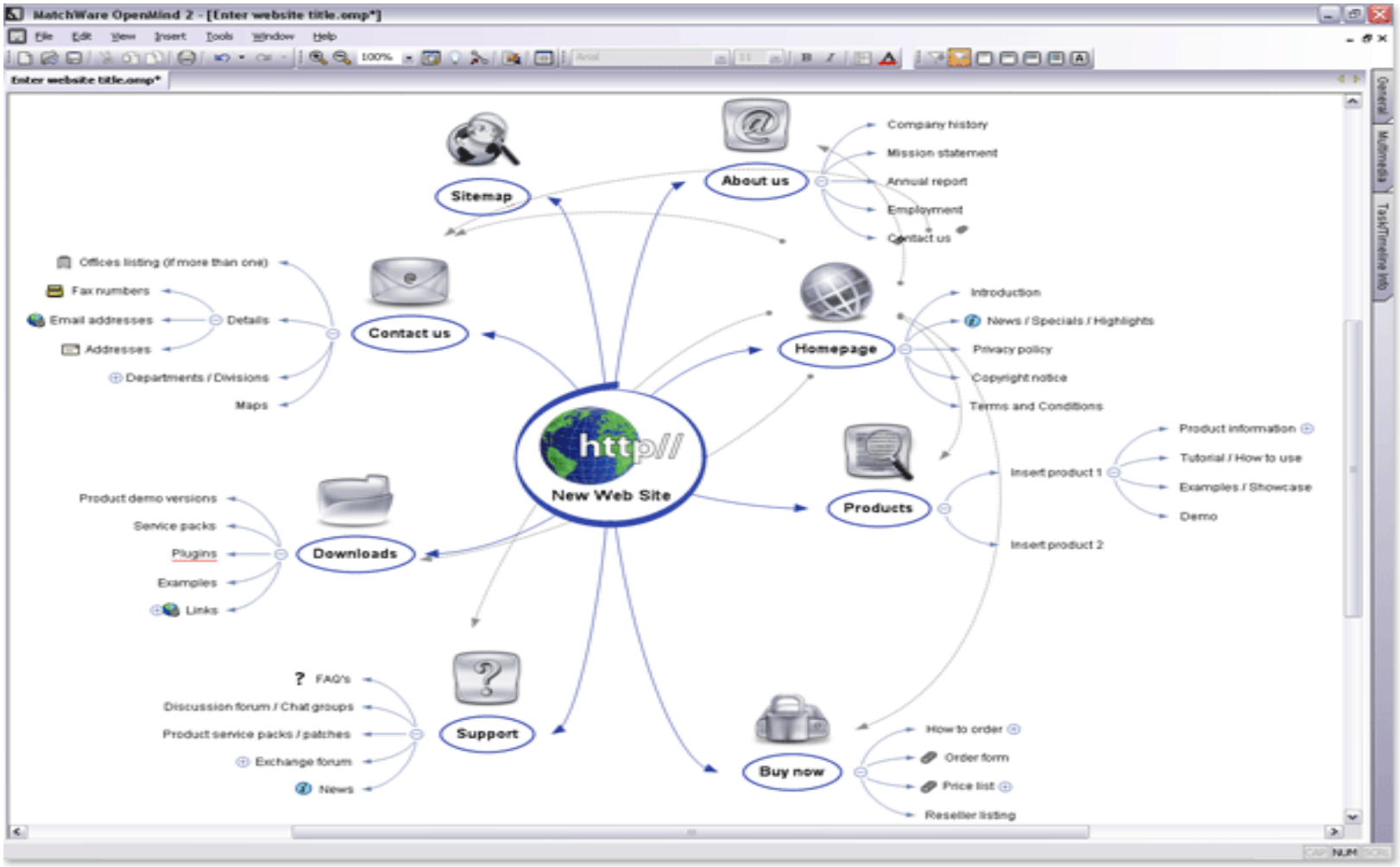
Audio: Sound of the motor running while the motors across the screen



Project Development Phases



Project Development Phases



Project Development Phases

- Specification and Design - Design - Processes
  - Designing the various **media elements** involves the detailed specification of type, number, format, quality and specific location of each media element of the product. Knowledge of the strengths and weaknesses of each medium is essential to ensure that their use is within the project's constraints and really contributing to the objectives of the product and not the reverse.
  - The **programming specifications** are written by developers and programmers to plan the programming processes that will integrate the interface, media components and functions into the final product.
  - If the **prototyping model** is used, the design document of the final product is only arrived at after several iterations, with each prototype being tested and evaluated to inform the next design. The prototype may be a representative 'slice' of the intended final product or it may focus on a particular component or function that needs to be clarified.

Project Development Phases

- Specification and Design - Design - Navigation and Interaction Plan
  - The basic navigational flow and interaction requirements of the package were already defined. The next task is to plot out the intricacies of the navigation and interaction. There are a range of ways in which this can be done, including flowcharts, storyboards, thumbnail sketches, diagrams, etc.
  - The flow chart and storyboard will indicate all the features of the product and how they interact. It will describe or indicate all of the buttons, icons, menus, navigation bars and shortcut options. Every element should be specified including forms, navigation elements, error and confirmation messages, etc.
    - For example, if a dialogue box is to appear when a button is pressed, the text to appear in the dialogue box will be noted here or in accompanying notes. If there are options to select from in the dialogue box, these will be identified and labeled, with a description of what happens when each is selected.

Project Development Phases

- Specification and Design - Design - Navigation and Interaction Plan
  - Detailing navigation and interaction can be a time consuming and sometimes confusing task.
  - Where you are developing a number of elements that will work together in a package you will need to design elements that work properly as stand-alone items and as well as in concert with the rest of the package. Every element becomes a test point when assessing the quality of the package.
  - Quality Assurance at this stage is very beneficial and can save a great deal of time and money later on. Each element should be assessed on its usability, compatibility and usefulness.

Project Development Phases

- Specification and Design - Design - Interface Design
  - Interface design involves combining the graphical treatment, content and interaction specifications into paper (or digital) mock-ups of the actual screens that the users will access to interact with these elements.
  - A range of issues must be considered and this includes (but is not limited to) the following:
    - Selection of fonts and styles - The selection of fonts can dramatically affect the readability of the package. It may be necessary to try a number of different fonts before selecting one that best suits the interface.
    - Screen layout - Good screen design can help to focus users' attention on the desired teaching points. It can have a significant impact on how they navigate through the package.

Project Development Phases

- Specification and Design - Design - Interface Design
  - A range of issues must be considered and this includes (but is not limited to) the following:
    - Selection of colors - The selection of colors can also affect the readability of the package. It may be necessary to try a number of different colors before selecting those that suit the interface.
    - Design of icons and buttons - The design of these elements has to contribute to the overall aesthetics and function of the package, as well as taking into account commonly accepted standards and cross-cultural factors.

Project Development Phases

- Specification and Design - Design - Media Elements
  - You have already determined, when scoping the package content, the types of media you require. These may include text, photographs and illustrations, animations, videos, narration, music and ambient sounds.
  - Designing the media elements will include the detailed specification of:
    - the type and number of each of these kinds of media
    - where they will be used in the product
    - what the required quality and format of those resources will be
    - who will be providing them and in what format, etc.

Project Development Phases

- Specification and Design - Design - Media Elements
  - In designing the media elements you will need to:
    - understand the strength and weaknesses of each type of medium in the context of the final product's objectives
    - be aware of the project's technical requirements and constraints that affect the use of the different types media
    - involve the experts/developers of each medium in determining the optimal use of that medium within the project's objectives and constraints
    - ensure that all media usage is really contributing to the product's objectives and is to the benefit of the users.
  - If the project design team already includes media expertise, the detailed treatment of those relevant elements may be determined.

Project Development Phases

- Specification and Design - Design - Media Elements
  - It may be necessary to use external providers to develop media. Some of these tasks can be quite complex (for example, in video production there are issues of lighting, sound and composition that must be considered, as well as editing, digitizing and compressing completed segments. Similarly, 3D modelling will possibly not be part of your team's skills repertoire).
  - If you are going to outsource media development it is particularly important to have specified the quality and performance characteristics you want for the media. These are the standards against which you will evaluate the media when they are provided to you by the contractor.

Project Development Phases

- Specification and Design - Design - Programming Specifications
  - The programming specifications are written by the project developers and programmers to plan the programming processes that they will use to integrate the interface, media components and the functions into the final product.
  - The team will also need to verify hardware and software requirements for developers and users against the programming specifications. Any changes to hardware and software requirements will need to be documented as per the project's change management procedures.

Project Development Phases

- Specification and Design - Design - Programming Specifications
  - Programming specifications are likely to identify:
    - the programming methodology and tools to be used
    - specific coding practices or languages to be used for specific sections of the package
    - code formatting rules
    - naming conventions and version control methods
    - any other standards and procedural approaches to be used in the implementation of the design
    - install and uninstall procedures
    - security
    - potential technological changes (for example, new versions of development software) during development cycle and their likely impact on the package.

Project Development Phases

- Specification and Design - Design - Programming Specifications
  - It is likely that this section will also include a list of performance characteristics against which the prototypes and final product will be evaluated, tested and quality managed. Those characteristics could include:
    - memory – amount of RAM required
    - speed – implications for machines and for networks (e.g. acceptable download times for networks, click response times, etc)
    - storage and online performance characteristics
    - visual feedback limitations
    - infrastructure characteristics (e.g. down time).

Project Development Phases

- Specification and Design - Proof of Concept
  - This phase of specification and design leads to the development of a detailed design document often accompanied by a proof of concept. You will need to sign off on the design document and provide feedback on the proof of concept. This is a significant **project milestone**.
  - The product design may still need to change in the next phase and this will be authorized and approved according to the agreed change control procedures. It is one of your important responsibilities throughout the project to keep to the agreed schedule for reviewing the deliverables. Delays will cause changes to the schedule and possibly the budget.

Project Development Phases

- Specification and Design - Proof of Concept
  - The proof of concept is an electronic 'mock-up' of the package. It demonstrates the design to both you and the team, allowing evaluation and validation of the design concept. Evaluation of the proof of concept may result in adjustments to the specification and design. The process results in the acceptance and sign off on the design as documented in the adjusted document.
  - Although the proof of concept is a valuable evaluation technique in the development process, the amount of time spent in its development should be limited. Only the key elements should be structured into the proof of concept.
  - It is essential to sign off on the design before proceeding with the full implementation of that design.

Project Development Phases

- Specification and Design - Proof of Concept
  - This process is supposed to demonstrate and validate the design and is not the actual start of development of the package.
  - Step One: The first step is to develop a 'mock up' of the interface that includes all the essential navigational features (buttons, icons, menus, etc) but not in their final polished form.
  - Step Two: The next step is to add a very small amount of content to the interface and a sample or two of the most crucial media. For example, if the package relies heavily on 3D models, a sample 3D model should be added with the main controls necessary to 'drive' it. If the package requires the display of mathematical symbols and tables, samples should be added.
  - Step Three: If you are happy with the mock-ups, then you will be asked to signoff on the design.

Project Development Phases

- Specification and Design - Proof of Concept - Evaluation and Validation
  - This is the process of seeking your feedback on the design demonstrated by the proof of concept and documented in the design document. Evaluation of the proof of concept may result in requests for adjustments to the specifications and design.
  - Sometimes, the design may be perceived as 'flawed' or inappropriate and the design team may have to 'go back to drawing board'. You may decide not to proceed or to go elsewhere! This could be a result of the design not being focused enough on your needs or your users' needs or some 'instability' in your needs. Sometimes this may be caused by changes in your organization in regards to who is making the project decisions.
  - Usually the process results in the acceptance and sign off on the design as documented in the design document. This is usually a significant milestone in the development lifecycle. The development phase of the package can then proceed.

- http://moodle.isep.ipp.pt
- cmc@isep.ipp.pt